

SUB-CHECK

Cloud-Based Behavioral Anti-Cheat Platform

Assignment 3: Customer Development (CustDev)

Course: Technology Entrepreneurship

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Repository: github.com/Scirtor/subcheck-anticheat

1. Problem Statement & CustDev Objectives

Why existing anti-cheat systems are failing the modern gaming ecosystem

Intrusive Kernel Drivers

Ring 0 rootkits (Vanguard, EAC) cause system instability, BSODs, and severe player privacy backlashes.

Hardware Cheats (DMA)

Second-PC PCIe DMA cards and hardware emulators bypass all client-side memory scanning entirely.

Indie & Linux Lockout

Enterprise tools cost 0k-0k+ upfront and lock out Linux, SteamOS, and Steam Deck players.

CustDev Objective: Validate these problem hypotheses with real players, tournament organizers, and indie developers, and quantify market demand for a server-side behavioral model.

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2. Audience Segmentation & Prioritization

Selecting key target segments based on accessibility, volume, and satisfaction

Segment	Profile & Description	Access	Volume	Impact	Priority
Segment A: Competitive Gamers	Ranked FPS players (CS2, Valorant, Apex, R6) suffering from cheaters & driver crashes.	5 / 5	5 / 5	4 / 5	P1 (Primary)
Segment B: Tournament Hosts / Admins	Collegiate leagues, community server admins battling closet cheaters & DMA hardware.	4 / 5	3 / 5	5 / 5	P2 (Secondary)
Segment C: Indie Game Developers	Studios needing affordable, easy-to-integrate multiplayer game integrity solutions.	3 / 5	3 / 5	5 / 5	P3 (B2B SaaS)

Execution: 32 structured questionnaire respondents + 5 in-depth recorded qualitative interviews conducted across Discord, Zoom, and Telegram.

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3. Key Quantitative Findings

Statistical feedback from 32 target market respondents

62.5% Negative towards Kernel-level software.

93.8% Positive toward Server-Side behavioral analytics.

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4. Key Customer Insights & Real Quotes

Crucial discoveries that reshaped our perspective

Tournament Organizer (R07): "Our biggest nightmare is DMA cards. A second PC reads game memory over PCIe hardware. No client-side anti-cheat catches it... A server-side behavioral model is the ONLY way."

Indie Studio Lead (R08): "Enterprise anti-cheats wanted 0k+ upfront licensing commitments. We were forced to launch unprotected. An indie-friendly SaaS model per active user is desperately needed."

Competitive Linux Gamer (R14): "I refuse to install rootkits on my work machine, and I was locked out of my favorite games on Linux. A server solution is completely OS-agnostic."

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5. Business Idea Updates (Post-CustDev)

How customer feedback directly adjusted our value proposition and go-to-market

Dimension	Initial Assumption	Validated Pivot
Customer Focus	B2C gamers via desktop app	B2B SaaS for Indie Studios & Tournament Organizers (direct purchasing power).

Dimension	Initial Assumption	Validated Pivot
Revenue Model	Flat annual enterprise license	Usage-based pricing (bash.03-bash.07 / MAU) with free tier for indie prototypes.
Key Differentiator	Generic anomaly heuristics	Kinematic jerk & reaction limit validation targeting DMA hardware & macros.
Enforcement	Automated binary banning	Continuous risk scoring (0.0-1.0) with quarantine matchmaking & admin dashboard.

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6. Conclusion & Next Milestones

Moving from Customer Validation to Technical Prototyping

- **CustDev Verdict: Strong Commercial & Technical Validation.** Market pain regarding kernel drivers and DMA hardware is at an all-time peak.
- **Milestone 1 (Month 1):** Ingestion SDK for game servers (Unreal Engine & Unity telemetry hooks).
- **Milestone 2 (Month 2):** Kinematic feature extraction model (human jerk physical limits, reaction latency).
- **Milestone 3 (Month 3):** Closed pilot with the 28 willing CustDev respondents and 2 collegiate esports leagues.

Thank You! Questions & Discussion

